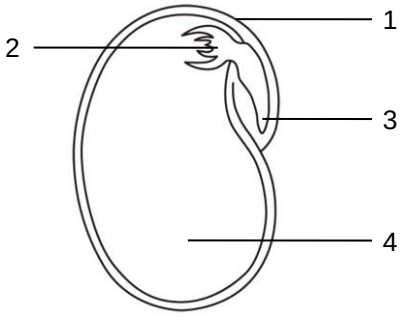
Multiple-choice questions

|!|BCEMC0150000001|!|

Directions: Questions (a) to (c) refer to the diagram below, which shows the longitudinal section of a bean seed.



(a) Which of the following is/are the component(s) of the embryo?

A. 4 only
B. 2 and 3 only
C. 2, 3 and 4 only
D. 1, 2, 3 and 4

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000002|!| ★

(b) Which of the following statements correctly describe the process of seed germination?

(1) Structure 1 prevents the absorption of excess water.
(2) Structure 3 is the first part that grows out through structure 1.
(3) The dry mass of structure 4 decreases.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

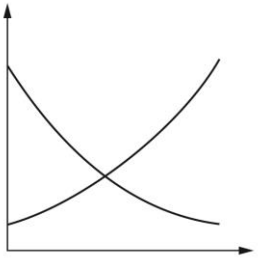
##C##

Skill assessed	Mark
Applying concepts	1

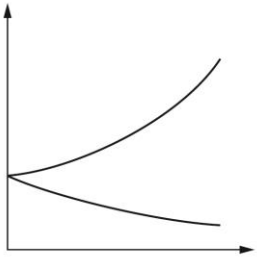
|!|BCEMC0150000003|!|

(c) Which of the following graphs correctly shows the changes in dry mass of structures 3 and 4 during germination?

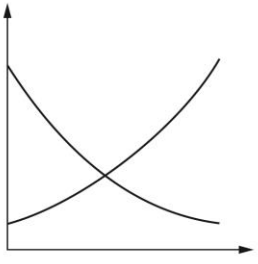
A.



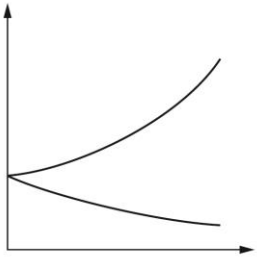
B.



C.



D.



☐

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000004|!| ★

Which of the following processes cause(s) the loss in dry mass of a germinating seed?

(1) respiration
(2) transpiration
(3) translocation

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

☐

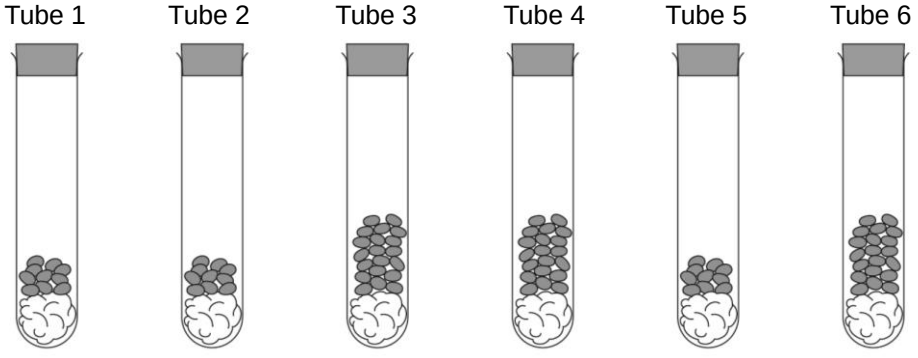
##A##

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Skill assessed	Mark
Applying concepts	1

|!|BCEMB0150000005|!| ★ SI

The diagram below shows a set-up used to investigate the conditions for seed germination.



Tube	1	2	3	4	5	6
Number of seeds	10	10	20	20	10	20
Condition of cotton wool	wet	wet	dry	wet	dry	dry
Temperature (°C)	5	20	5	20	5	20

Which of the groups of tubes can be compared to show the effect of temperature on seed germination?

(1) 1 and 2
(2) 4 and 5
(3) 3 and 6

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

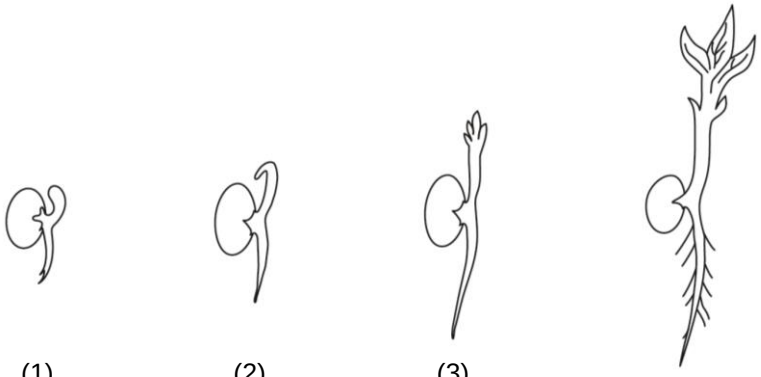
□

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0150000006|!|

The diagram below shows various stages in the germination of a broad bean seed.



The seedling starts to increase in dry mass at stage

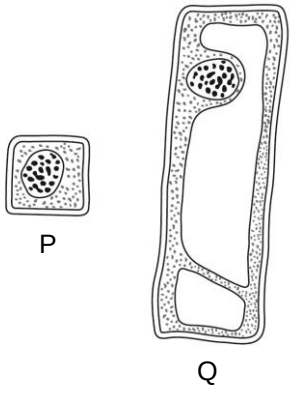
A. (1).
B. (2).
C. (3).
D. (4).

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000007|!| ★

The two cells P and Q below can be observed in the young root of a seedling.



Which of the following statements about the two cells is/are correct?

(1) Cell P develops from cell Q.
(2) Cell P is nearer the root tip than cell Q.
(3) Cell P will never have vacuole.

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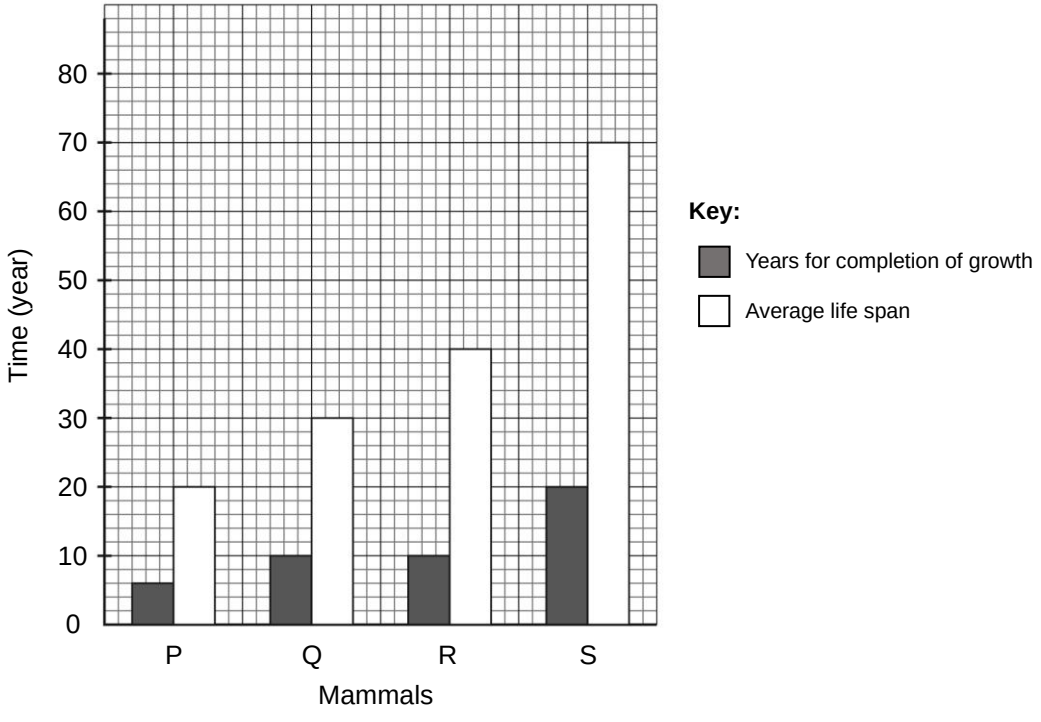
A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000008|!| ★

The bar chart below shows a comparison of the growth-time and life span of four mammals P, Q, R and S.



Which of the following mammals spends the least part of its life in growing?

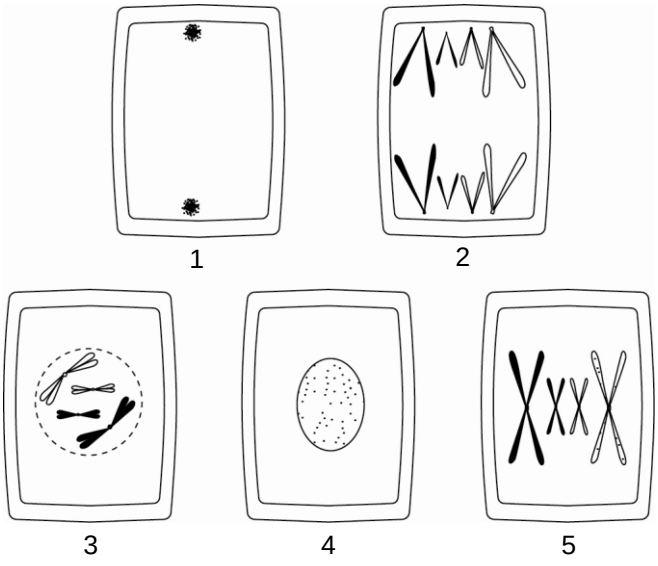
A. P
B. Q
C. R
D. S

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000009|!| ★ **Cross topic**

Directions: Questions (a) and (b) refer to the diagram below which shows the stages of the mitotic cell division in a cell.



(a) The correct sequence is

A. 1 → 5 → 3 → 2 → 4.

B. 3 → 4 → 5 → 2 → 1.

C. 4 → 1 → 2 → 5 → 3.

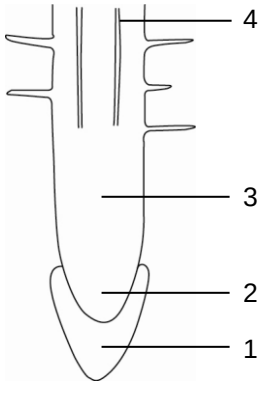
D. 4 → 3 → 5 → 2 → 1.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000010|!| ★

(b) The following diagram shows the vertical section of the root tip of an onion plant.



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In which region of the root tip would the mitotic cell division shown in (a) be found?

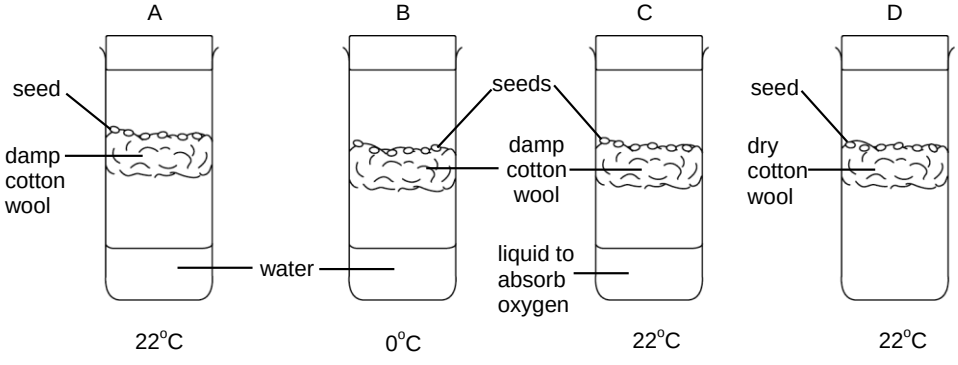
A. 1
B. 2
C. 3
D. 4

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000011|!| ★ **SI**

The diagram below shows the set-ups used to investigate the requirements for seed germination.



In which test tube would the seeds germinate?

A. test tube A
B. test tube B
C. test tube C
D. test tube D

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0150000012|!|

Which of the following are the characteristics of the growth in humans?

(1) The growth of a person changes with age.
(2) Males and females have different growth rates during puberty.
(3) Different parts of the body grow at different rates.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000013|!|

Which of the following are the functions of seed coats?

(1) to protect the embryo
(2) to prevent bacterial infection
(3) to carry out photosynthesis

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000014|!|

The embryo in a seed is made up of

(1) a plumule.
(2) a radicle.
(3) cotyledon(s).

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000015|!|
Cotyledons in a dicotyledonous seed contain

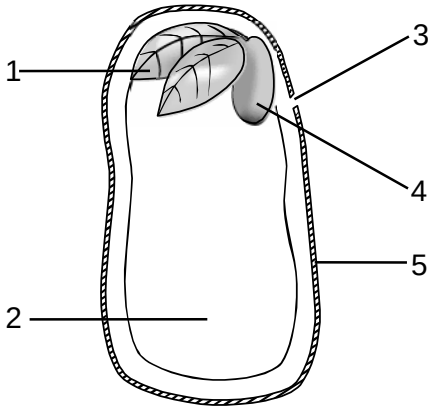
(1) proteins.
(2) starch.
(3) lipids.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000016|!|
Directions: Questions (a) and (b) refer to the following diagram which shows a section of a seed.



(a) Which of the following combinations correctly identifies structures 2 and 4?

	Structure 2	***Structure 4***
A.	plumule	radicle
B.	radicle	plumule
C.	cotyledon	radicle
D.	radicle	cotyledon

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000017|!| ★

(b) Which of the following statements about the seed is/are correct?

(1) The embryo includes structures 1, 2, 4 and 5.
(2) The energy for germination comes from structure 1.
(3) Structure 5 is developed from the integument of parent plant.

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

□

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0150000018|!|

Which of the following are necessary for seed germination?

(1) water
(2) sunlight
(3) oxygen

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##B##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000019|!|

Which of the following statements about seed coat is ***incorrect***?

A. Seed coat will rupture when cotyledon absorbs water.

B. Seed coat is developed from the ovary wall of the parent plant.

C. Hilum on the seed coat is formed when the seed is detached from the ovary.

D. Seed coat protects the embryo against fungal attack.

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000020|!|

Which of the following about the components of seeds is ***incorrect***?

A. A micropyle is a pore through which water and oxygen enter the seed.

B. A radicle will develop into a root.

C. A cotyledon will develop into a shoot.

D. Seed coat is developed from the integument of the parent plant.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000021|!|

Which of the following does ***not*** occur during seed germination?

A. Green leaves are developed from the cotyledons.

B. Cells in the growing point divide actively by mitotic cell division.

C. The embryo increases in dry mass.

D. Enzymes are activated to break down food reserves in the cotyledons.

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0150000022|!|

At the root tip of a plant, which of the following regions has / have cells with no vacuoles?

(1) region of elongation
(2) region of differentiation
(3) region of cell division

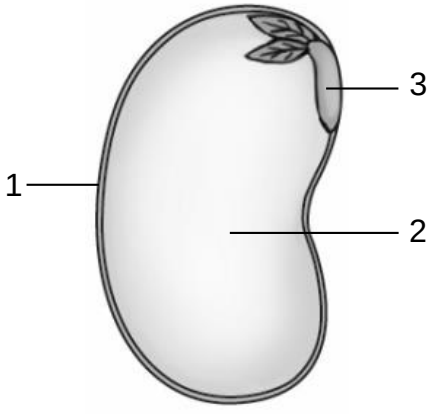
A. (2) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000023|!|

Directions: Questions (a) and (b) refer to the diagram below, which shows the longitudinal section of a seed.



(a) Which of the following combinations correctly identifies structures 1 to 3?

	Structure 1	***Structure 2***	***Structure 3***
A.	epidermis	cotyledon	radicle
B.	epidermis	embryo	plumule
C.	seed coat	cotyledon	radicle
D.	seed coat	embryo	plumule

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0150000024|!|

(b) Which of the following structures will shrink and fall off from the plant after germination?

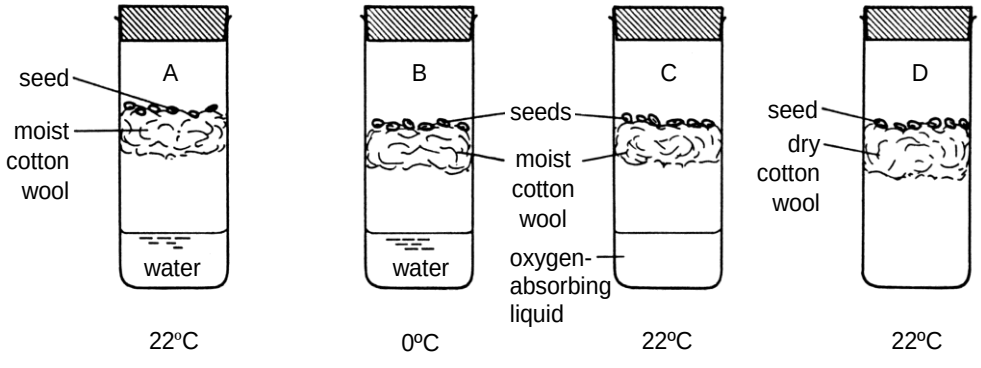
A. 1 only
B. 3 only
C. 1 and 2 only
D. 2 and 3 only

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0150000025|!| **SI**

The following diagram shows the experimental set-ups for investigating the conditions necessary for seed germination.



The percentage of seed germination is the highest in

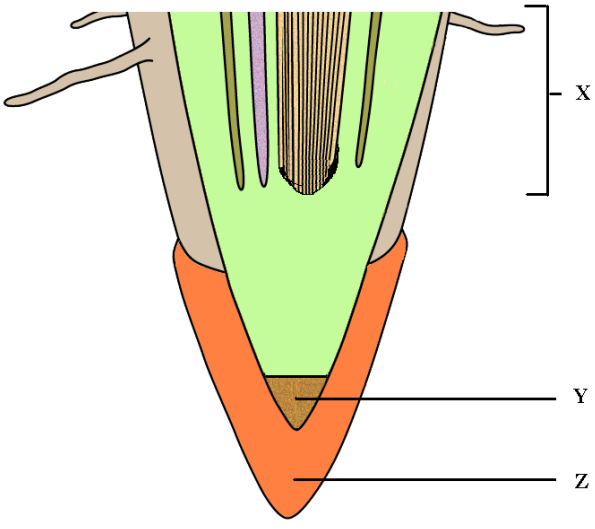
A. test tube A.
B. test tube B.
C. test tube C.
D. test tube D.

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0150000026|!|

The diagram below shows the longitudinal section of a young root.



Which of the following statements about the cells in region X are correct?

(1) The cells have no central vacuoles.
(2) The cells become specialized to carry out different functions.
(3) The sizes of the cells are larger than those in region Y.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000027|!| ★

Which of the following is/are ***not*** example(s) of development?

(1) formation of blood cells from the bone marrow
(2) increase in the size of a root cell after water absorption
(3) emergence of root hairs from the root epidermal cell

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

□

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0150000028|!|

During the first two days of seed germination, which of the following increase(s) in content?

(1) water
(2) starch
(3) amino acids

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

□

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0150000029|!|

What is/are the disadvantage(s) of measuring the growth of an elephant by measuring its dry mass?

(1) This method is inaccurate.
(2) The elephant must be killed when using this method.
(3) This method is time-consuming.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000030|!|

Which of the following is/are responsible for the secondary growth in plants?

(1) increase in the length of the root
(2) increase in the height of the plant
(3) increase in the girth of the plant

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0150000031|!|

Which of the following statements about growth is/are correct?

(1) Growth is the irreversible increase in the size of an organism.
(2) Growth occurs by meiotic cell division.
(3) Growth involves the differentiation of cells.

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000032|!|

Which of the following is/are related to growth?

(1) increase in size of cells
(2) increase in dry mass of cells
(3) increase in number of cells

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0150000033|!| ★

Which of the following growth parameters is the ***most*** suitable for measuring the growth of a tree?

A. total surface area of the leaves
B. dry mass of the tree
C. length of the stem
D. fresh mass of the tree

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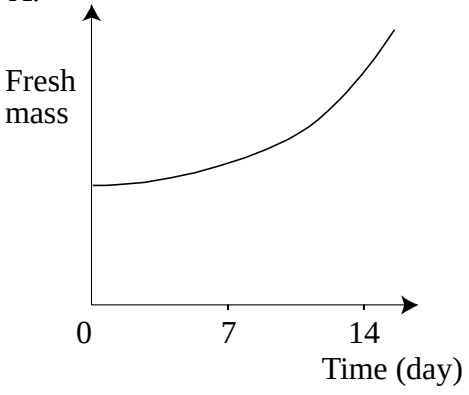
##C##

Skill assessed	Mark
Applying concepts	1

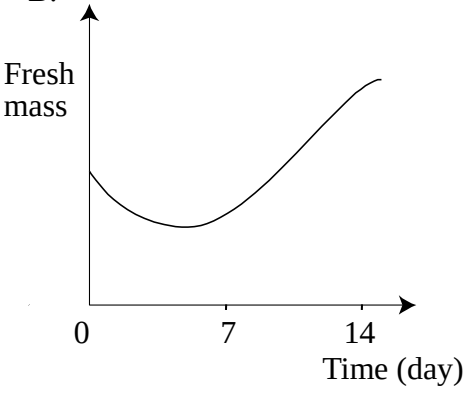
|!|BCEMB0150000034|!| ★

Which of the following graphs shows the changes in the fresh mass of a germinating seeds during 14 days after germination?

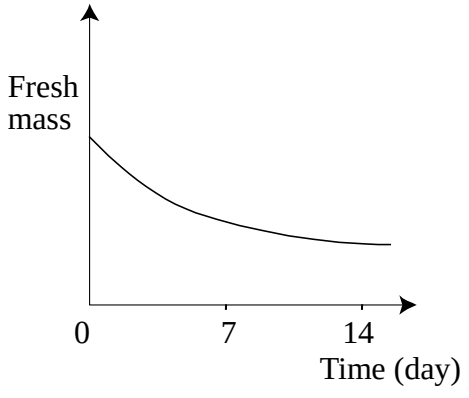
A.



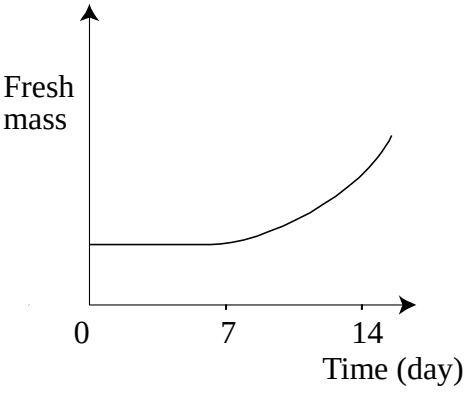
B.



C.



D.



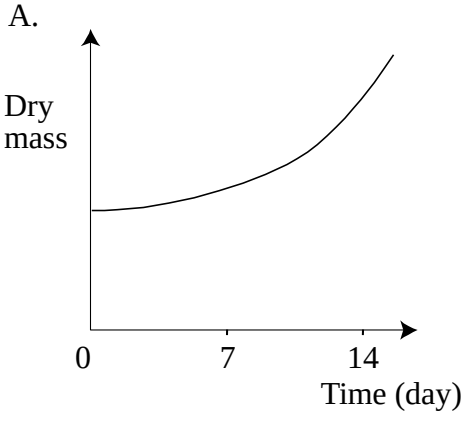
##A##

Skill assessed	Mark
Analysing data	1

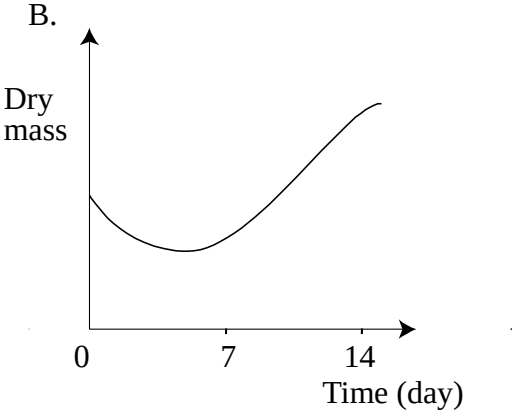
|!|BCEMB0150000035|!| ★

Which of the following graphs shows the changes in the dry mass of a germinating seeds during 14 days after germination?

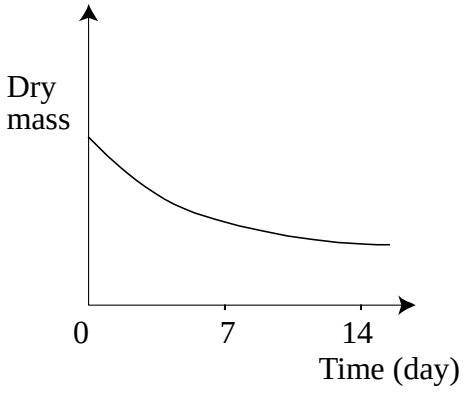
A.



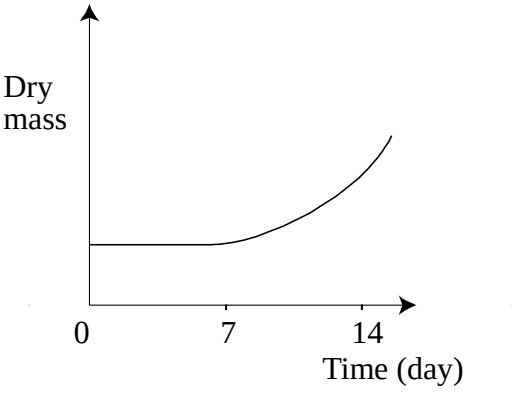
B.



C.



D.



##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0150000036|!|

Which of the following is ***not*** responsible for the increase in the thickness of the stem of flowering plants?

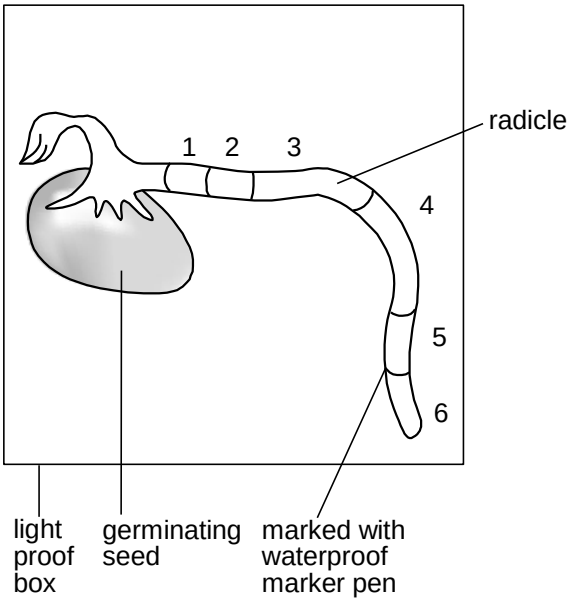
A. mitotic cell division in apical meristem
B. growth of secondary phloem
C. growth of secondary xylem
D. mitotic cell division in vascular cambium

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000037|!| **SI**

In an investigation, a student first used waterproof marker pen to mark six lines at regular intervals on the radicle of a germinating seed. After two days, the appearance of the germinating seed is shown below.



To which part of the root do zones 3 and 4 belong?

A. region of cell division
B. region of elongation
C. region of differentiation
D. root cap

□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000038|!| ★

A boy carves a letter ‘A’ on the bark of a tree (1 m tall) at a height of 0.5 m above the ground. After five years, the tree doubles its height. At what height from the ground would the letter be found on the tree trunk?

A. 0.5 m above the ground
B. 1 m above the ground
C. 1.5 m above the ground
D. 2 m above the ground

□

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##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0150000039|!|

Which of the following statements about plant growth are correct?

(1) The increase in length is due to the growth in apical meristem.
(2) Growth stops after plants are mature.
(3) The formation of vacuoles causes further growth of plant cells.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

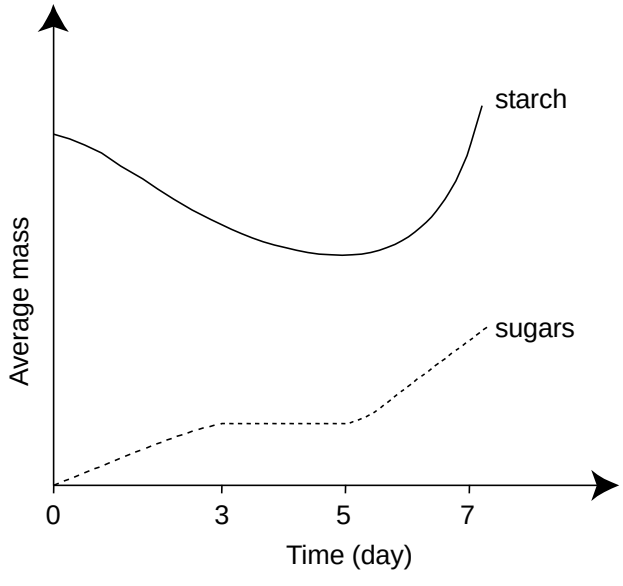
□

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0150000040|!| ★

Directions: Questions (a) to (c) refer to the following graph which shows the changes in the mass of starch and sugars of a seed after germination, in a period of a week.



(a) Which of the following can explain the rise of sugar level before day 5?

A. The plant carried out photosynthesis to make sugars.
B. Soil microorganisms made sugars for the plant.
C. Starch in seed was broken down to form sugars.
D. Sugars were absorbed from the soil.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000041|!| ★

(b) Which of the following is/are the function(s) of sugars before day 5?

(1) They are broken down to release energy by respiration.
(2) They are used to form new cells.
(3) They are converted to starch for storage.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0150000042|!| ★

(c) Which of the following statements about the germinating seed is/are correct?

(1) No sugar was formed from day 3 to day 5.
(2) After day 5, green leaves were developed.
(3) The mass of starch rose after day 5 because the rate of photosynthesis exceeded that of respiration during daytime.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

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Skill assessed	Mark
Applying concepts	1

|!|BCEMC0150000043|!|
Which of the following is/are the function(s) of water in seed germination?

(1) to soften the seed coat
(2) to activate enzymes
(3) to hydrolyse food

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000044|!|
Which of the following is the first part of a seed to emerge during germination?

A. plumule
B. radicle
C. cotyledon
D. seed coat

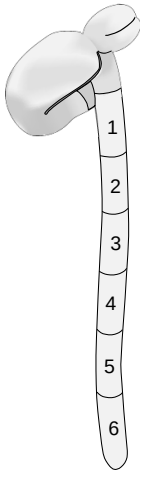
##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0150000045|!| **SI**

Directions: Questions (a) and (b) refer to the following diagram which shows the growth of the radicle from a seed. The radicle was marked with six equal intervals with a waterproof pen.



(a) Which region will show the greatest change in length after several days?

A. 1

B. 2

C. 4

D. 6

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0150000046|!| **SI**

(b) Meristematic cells are located in region

A. 1.

B. 2.

C. 3.

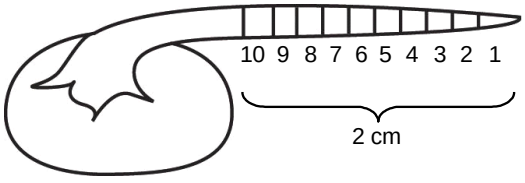
D. 6.

##D##

Skill assessed	Mark
Analysing data	1

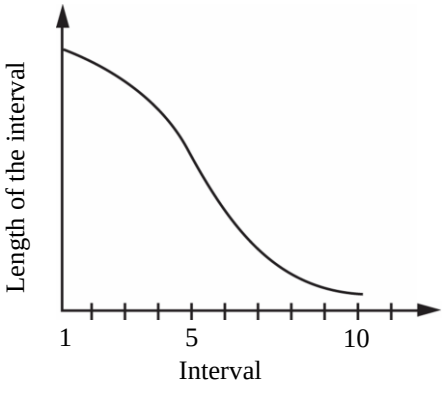
|!|BCEMB0150000047|!| ★ SI

The following diagram shows the radicle of a young broad bean seedling which is marked with ten lines at regular intervals, using a fadeproof marker pen.

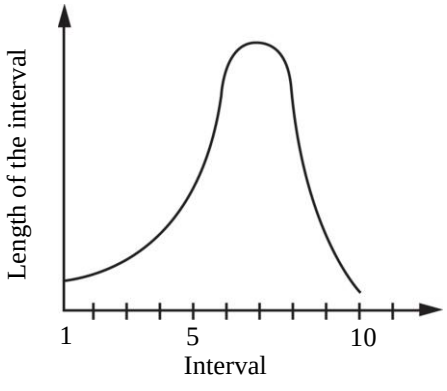


Which of the following graphs shows the lengths of the different intervals after two days?

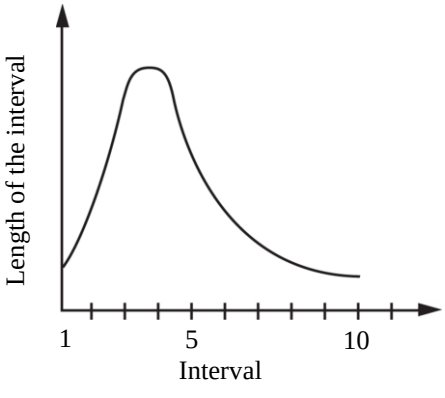
A.



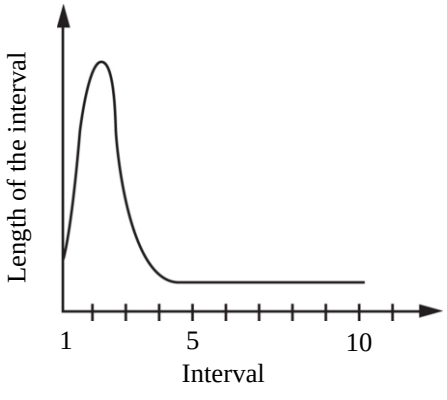
B.



C.



D.

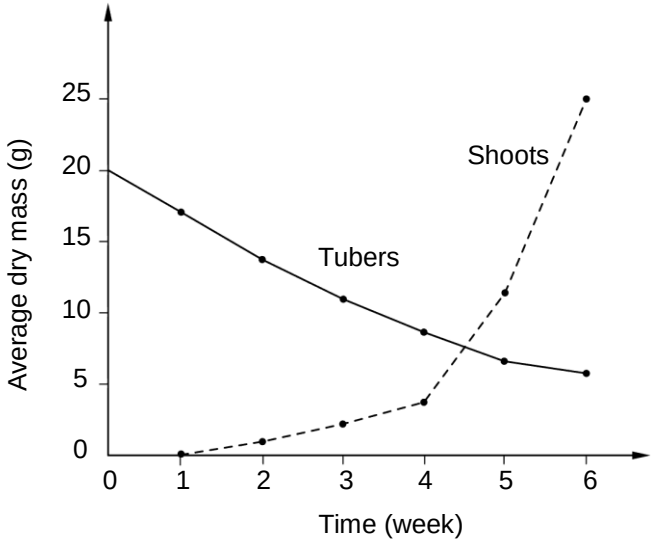


##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000048|!| ★

Directions: Questions (a) to (c) refer to the graph below which shows the average dry mass of potato tubers and the shoots produced from the tubers during a period of six weeks.



(a) The decrease in average dry mass of the tubers by the end of the six weeks is

A. 5 g.
B. 6 g.
C. 14 g.
D. 20 g.

##C##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMB0150000049|!| ★

(b) Which of the following are the reasons for the decrease in the dry mass of the potato tubers?

(1) Some of the food stored in the tubers has been used for the growth of new tubers.
(2) Some of the food stored in the tubers has been used for the shoot growth.
(3) Some of the food stored in the tubers has been used for the root growth.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

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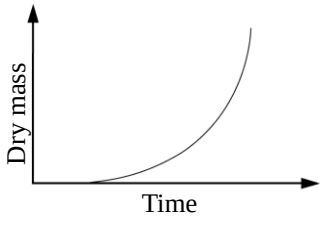
##C##

Skill assessed	Mark
Applying concepts	1

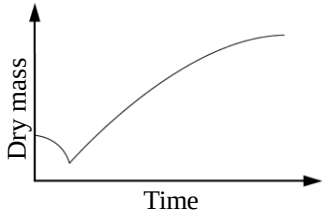
|!|BCEMB0150000050|!| ★

(c) Which of the following graphs ***best*** represents the changes in the dry mass of the whole potato plant including the tubers during the growth period?

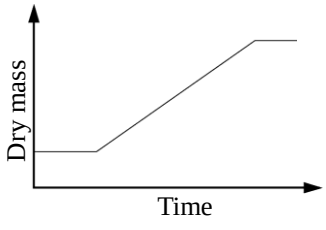
A.



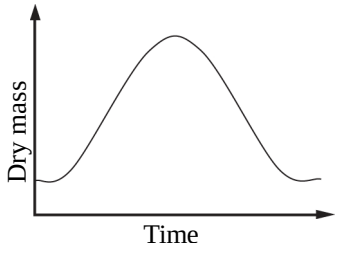
B.



C.



D.



##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0150000051|!|

Which of the following statements about the growth rates of different parts of the human body is ***incorrect***?

A. The arms and legs grow rapidly during childhood.

B. The reproductive organs grow rapidly during puberty.

C. The relative size of the body trunk remains more or less the same at different ages.

D. The relative sizes of the head and brain decrease with age.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000052|!|

Which of the following is/are the advantage(s) of using fresh mass to measure the growth of an organism, when compared with dry mass measurement?

(1) There is no need to kill the organism.
(2) Continuous measurement can be taken for the same organism.
(3) It is more accurate.

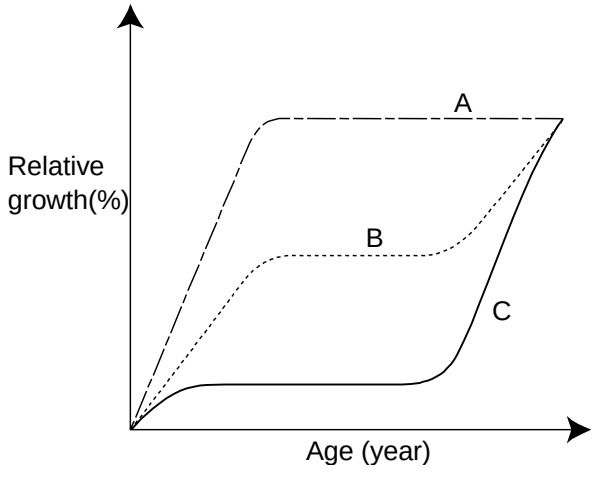
A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0150000053|!|

The following graph shows the relative growth of different parts of the human body.



Which of the following combinations correctly matches the curves with the corresponding body parts?

	A	***B***	***C***
A.	reproductive organs	brain and skull	whole body
B.	whole body	reproductive organs	brain and skull
C.	brain and skull	whole body	reproductive organs
D.	whole body	brain and skull	reproductive organs

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0150000054|!|

The following shows the processes involved in the growth and development in plants.

(1) mitosis
(2) cell elongation
(3) cell differentiation
(4) cell plate formation

Which of the following is the correct order of the processes?

A. (1) → (2) → (3) → (4)
B. (1) → (4) → (2) → (3)
C. (4) → (1) → (2) → (3)
D. (4) → (2) → (1) → (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0150000055|!| ★

Which of the following statements about the growth and development of plants is/are ***incorrect***?

(1) The dry mass of a plant increases proportionally with time.
(2) Secondary growth involves the increase in the length of the plant.
(3) Increase in the length of a plant occurs mainly in the region of elongation.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0150000056|!| ★

Which of the following statements about the differentiation of cells in xylem tissues is/are correct?

(1) The end walls of the cells are broken down.
(2) The cell walls are deposited with lignin.
(3) The cytoplasm is removed.

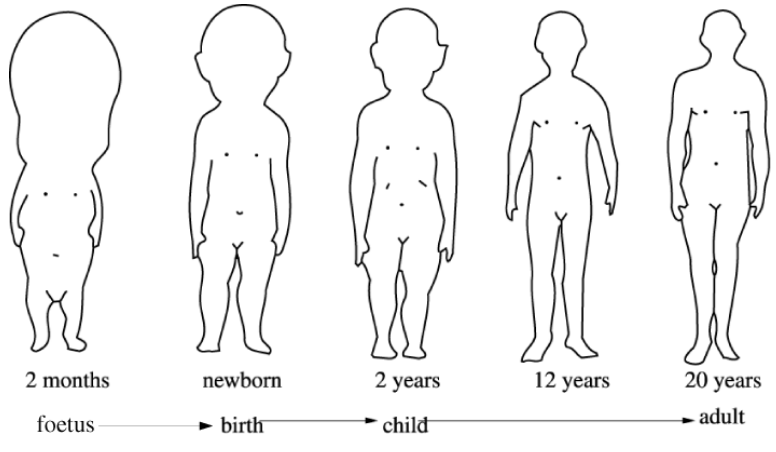
A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0150000057|!| ★

The diagram below shows the stages of growth in the life of a man. All the individuals are drawn to the same height for easy comparison.



Which of the following statements about the growth of the man during the 20 years is ***incorrect***?

A. The relative length of legs increases with age.
B. Different parts of the body develop at different rates.
C. The relative size of the body trunk remains more or less the same.
D. The size of the head decreases with age.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000058|!| ★

The following table shows the relative sizes of the head and the body of humans at different ages. The head size is fixed at the value of '1' while the body size relative to the head size is given by the number in the table.

Age	Head size	Body size
Newborn	1	4
2 years	1	5
12 years	1	6
Adult	1	8

Which of the following conclusion about the table is correct?

A. The head is not growing all the time.
B. The relative size of the body is different at different times.
C. The growth of the body is the fastest from newborn to age 2.
D. The body size grows faster than the head.

□

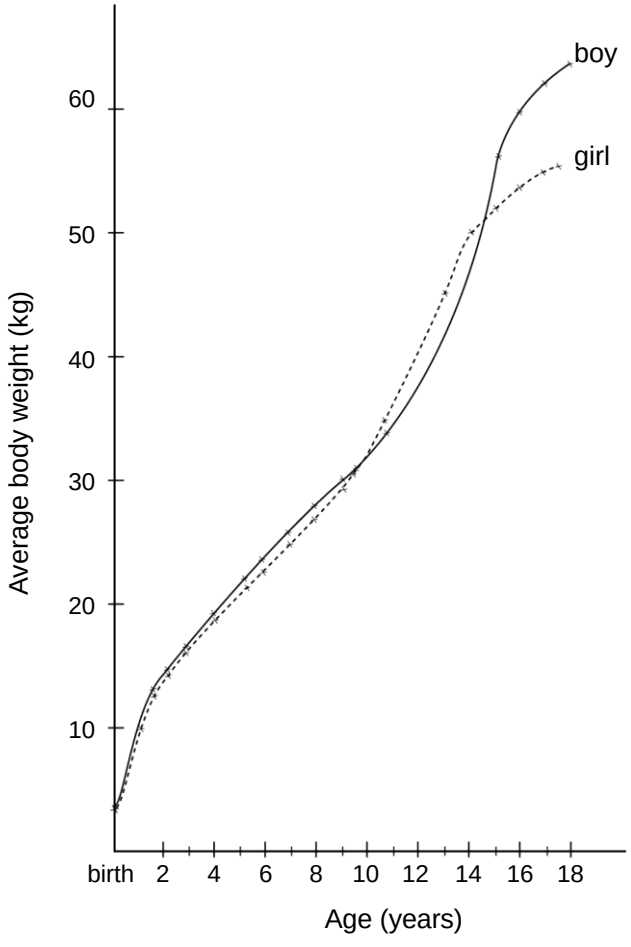
##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0150000059|!| ★

The graph below shows the average body weights of boys and girls at different ages.



Which of the following statements about the growth of boys and girls is ***incorrect***?

A. The growth rates of newborn boys and girls are the same.

B. The growth rates of boys and girls before puberty are higher than that during puberty.

C. Between age 11 and 14, the growth rate is higher in girls than in boys.

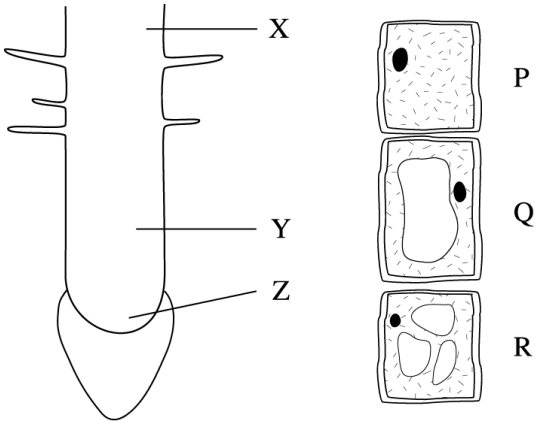
D. Between age 16 and 18, the growth rate is higher in boys than in girls.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0150000060|!| ★

The diagram below represents a root tip and three cells taken from regions X, Y and Z.



Which of the following combinations correctly matches cells P, Q and R with their corresponding regions?

	Region X	***Region Y***	***Region Z***
A.	Cell P	Cell R	Cell Q
B.	Cell Q	Cell P	Cell R
C.	Cell Q	Cell R	Cell P
D.	Cell R	Cell Q	Cell P

##C##

Skill assessed	Mark
Analysing data	1